Retrieval Evaluation and Query operations

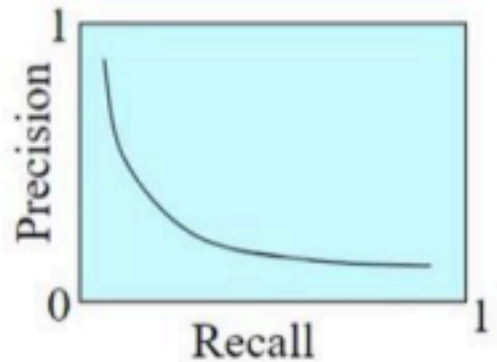
Module 2

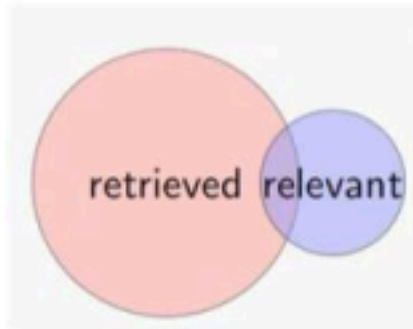
Precision (P) is the fraction of retrieved documents that are relevant.

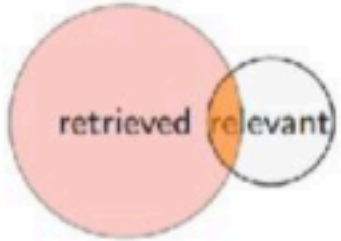
Recall (R) is the fraction of relevant documents that are retrieved

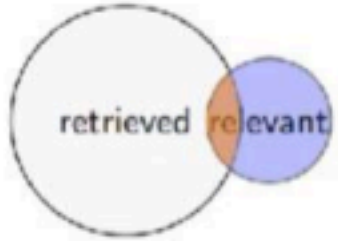
$$\text{Precision} = \frac{\#(\text{relevant items retrieved})}{\#(\text{retrieved items})}$$

$$\text{Recall} = \frac{\#(\text{relevant items retrieved})}{\#(\text{relevant items})}$$









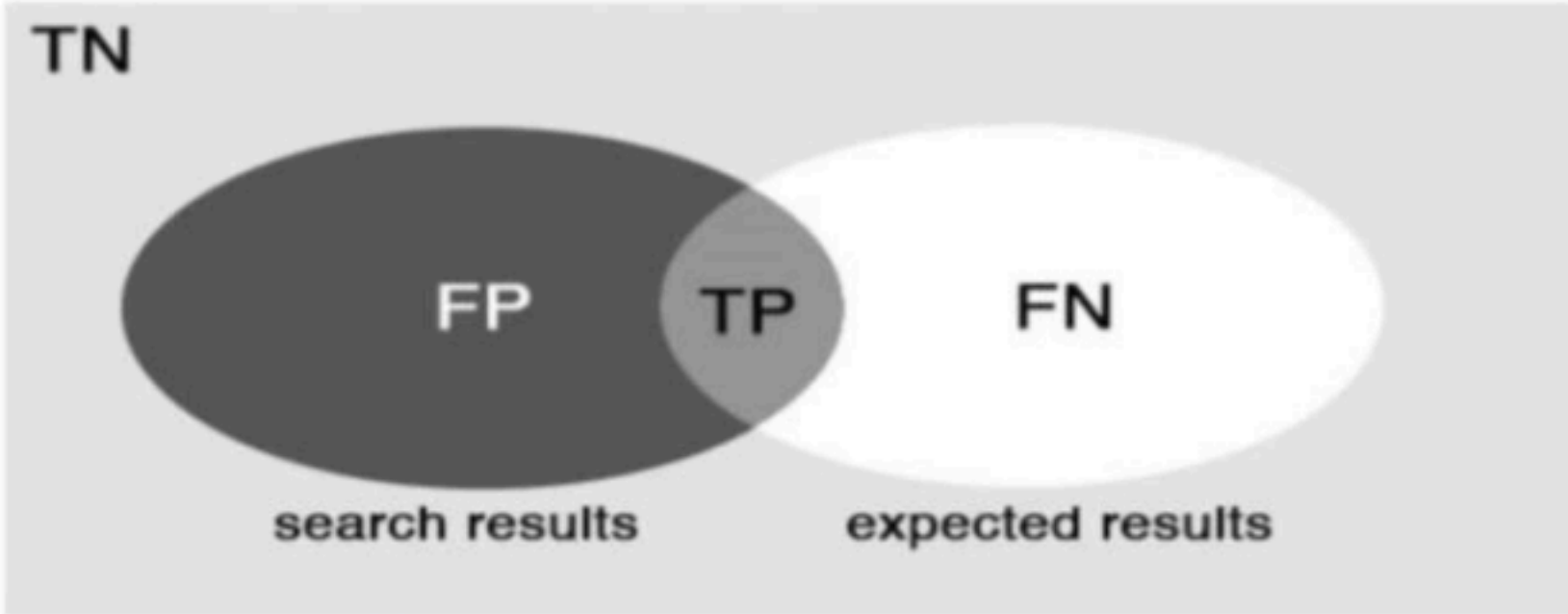
	Actual		
Predicted		Relevant	Non Relevant
	Retrieved	True Positive (TP)	False Postive (FP)
	Not Retrieved	False Negative(FN)	True Negative (TN)

$$\text{Precision}(P) = \frac{TP}{TP + FP}$$

$$\text{Recall}(R) = \frac{TP}{TP + FN}$$

Recall = 1 when FN=0, since 100% of the TP were discovered

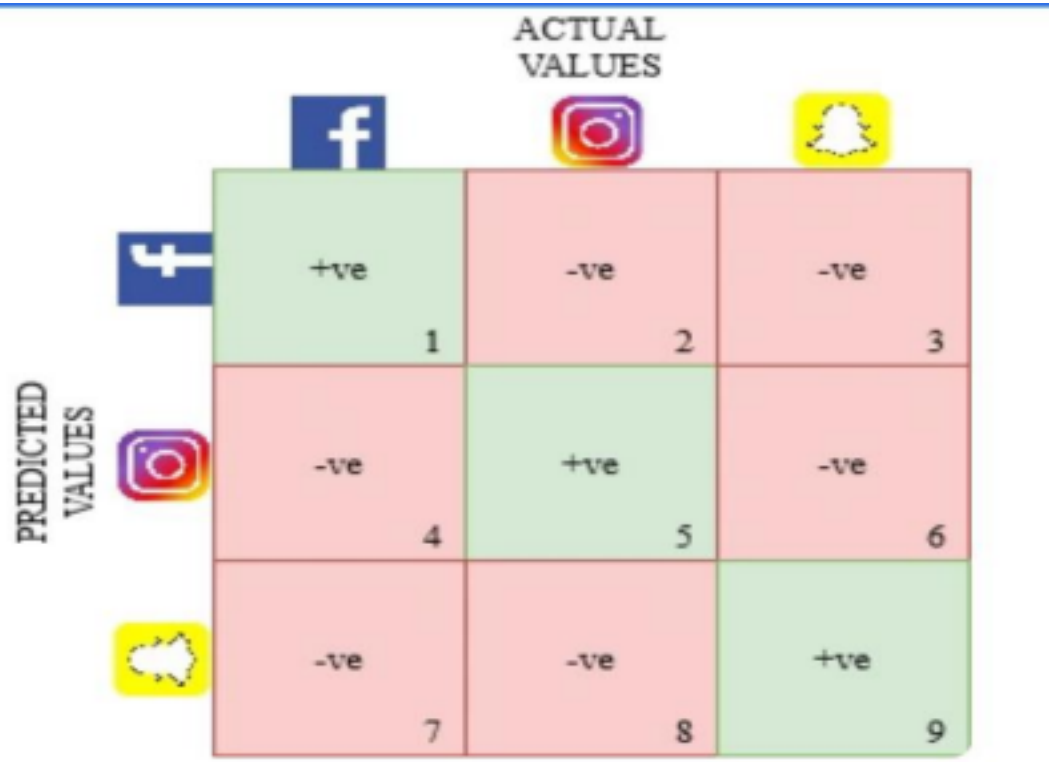
Precision = 1 when FP=0, since no there were no spurious results



The Venn diagram of the confusion matrix

	Actually Pregnant	Actually NOT Pregnant
Predicted Pregnant	 True Positive (TP)	 False Positive (FP)
Predicted NOT Pregnant	 False Negative (FN)	 True Negative (FN)

Confusion Matrix



Facebook	Instagram	Snapchat
TP = $Cell_1$	TP = $Cell_5$	TP = $Cell_9$
FP = $Cell_2 + Cell_3$	FP = $Cell_4 + Cell_6$	FP = $Cell_7 + Cell_8$
TN = $Cell_5 + Cell_6 + Cell_8 + Cell_9$	TN = $Cell_1 + Cell_3 + Cell_7 + Cell_9$	TN = $Cell_1 + Cell_2 + Cell_4 + Cell_5$
FN = $Cell_4 + Cell_7$	FN = $Cell_2 + Cell_8$	FN = $Cell_3 + Cell_6$

Example 1: Total 16 docs are there. For a given query, **8 are relevant** and **8 are non relevant.**

For a given query, system retrieves 10 documents. 4 are relevant and 6 are non relevant

1.	d1	yes
2.	d2	yes
3.	d3	no
4.	d4	yes
5.	d5	no
6.	d6	no
7.	d7	no
8.	d8	no
9.	d9	no
10.	d10	yes

		Actual	
		Relevant (8)	Non Relevant (8)
Predicted	Retrieved (10)	TP(4)	FP(6)
	Not Retrieved	FN(4)	TN(2)

$$\text{Precision(P)} = \frac{TP}{TP + FP} \qquad \text{Recall(R)} = \frac{TP}{TP + FN}$$

$$\text{Precision(P)} = \frac{4}{4+6} = \frac{4}{10} = 0.4 \qquad \text{Recall(R)} = \frac{4}{4+4} = \frac{4}{8} = 0.5$$

Example 2: Documents available (10): D1,D2,D3, D4, D5, D6, D7, D8, D9,D10

Relevant to our need (5): D1, D4, D5, D8, D10

Query to search engine retrieves(6): D2, D4, D5, D6, D8, D9

No.	Doc. No.	Relevant
1	D2	No
2	D4	Yes
3	D5	Yes
4	D6	No
5	D8	Yes
6	D9	No

		Actual	
		Relevant (5)	Non Relevant (5)
Predicted	Retrieved (6)	TP(3) [D4,D5, D8]	FP(3) [D2,D6,D9]
	Not Retrieved (4)	FN(2) [D1,D10]	TN(2) [D3,D7]

$$\text{Precision}(P) = \frac{3}{3+3} = \frac{3}{6} = 0.5$$

$$\text{Recall}(R) = \frac{3}{3+2} = \frac{3}{5} = 0.6$$

Example 7: An IR system returns 12 relevant documents and 10 irrelevant documents. There are a total of 25 relevant documents in the collection. What is the precision of the system on this search and what is the recall.

		Actual	
		Relevant (25)	**Non Relevant**
Predicted	**Retrieved (22)**	**TP(12)**	**FP(10)**
	Not Retrieved	**FN(13)**	**TN**

$$\text{Precision(P)} = \frac{12}{12+10} = \frac{12}{22} = 0.55$$

$$\text{Recall(R)} = \frac{12}{12+13} = \frac{12}{25} = 0.48$$

- Precision can be seen as **a measure of quality,** and recall as **a measure of quantity.**
- **Higher precision means that an algorithm returns more relevant results than irrelevant ones, and high recall means that an algorithm returns most of the relevant results.**
- A high value of **precision means a large part of retrieved documents are relevant.**
- Web surfers would like **every result on the first page to be relevant (high precision)** but users have not the slightest interest in knowing alone looking at every document that is relevant
- A high value of **recall means a large part of relevant documents are retrieved**
- Professional searchers (paralegals and intelligence analysts)are very concerned with **trying to get as high recall as possible**

Accuracy

The fraction of correctly classified documents is ontained by Accuracy

$$\text{Accuracy} = \frac{TP + TN}{N}$$

where N is the total number of N documents in the collection

- In reality, most of the documents (99.9%) are non-relevant, so it would be relatively easy to find non-relevant documents
- If an IR system is tuned to maximize its accuracy, it is likely to classify all documents as non-relevant
- It means the system will have a high TN
- Users want to **see some documents (TP+FP) with a relatively large TP and small FP**
- Therefore precision, recall and F measure are used for evaluation rather than Accuracy

F1-score: **Simple average of Precision & Recall** is bad idea

F1-score: **Weighted average (Harmonic mean)** is good idea

Bad idea

$$F1 - Score = \frac{Precision + Recall}{2}$$

Good idea

$$F1 - Score = 2 \times \frac{Precision * Recall}{Precision + Recall}$$

Classifier	Precision	Recall	F1-Score (Simple Avg)	F1-Score (Harmonic Mean)
1	0.5	0.5	0.5	0.5
2	1	0.0001	0.50005	0.0002

Reference collection

- Reference collection used throughout the year for the evaluation of information retrieval system.
- **Background & Need for TREC**
- Information Retrieval (IR) research was criticized

  for: • Lacking a solid formal framework.
  • Lacking robust and consistent testbeds and benchmarks.
- Small, inconsistent test collections made it hard to compare

systems. • Subjectivity in relevance judgments limited progress.

- **Creation of TREC**
- Initiated in the early 1990s under **Donna Harman** at **NIST** (National Institute of Standards and Technology).
- Supported by **DARPA's TIPSTER Text Program**.

- Aimed to:
  - Provide large-scale test collections (millions of documents).
  - Establish uniform evaluation procedures.
  - Offer a platform for comparing retrieval systems.
- **TREC Conferences**
- First held in **1992** at NIST; annual conferences since then.
- Participants: universities, research centers, corporations (e.g., Apple, AT&T Labs, IBM, Carnegie Mellon, Xerox, etc.).
- Systems evaluated using:
  - Large test collections (initially 2GB ⟶ later ~6GB, >1 million documents). • Standardized queries (called **topics**).
  - Common evaluation packages for fair comparison.

- **Document Collection**
- Composed of documents + topics + relevance judgments.
- Distributed on **six CD-ROMs** (~1GB each).
- Sources include:
  - **WSJ (Wall Street Journal)**

  - **AP (Associated Press newswire)**
  - **ZIFF (Computer Selects, Ziff-Davis)**
  - **FR (Federal Register)**
  - **DOE Publications**
  - **San Jose Mercury News**
  - **US Patents**
  - **Financial Times**
  - **Congressional Record**
  - **FBIS (Foreign Broadcast Information Service)**
  - **LA Times**
- Documents encoded in **SGML** for easy parsing, with fields like <DOCNO> (document number) and <TEXT> (content).
- Example statistics (TREC-6): ~5.8 GB, over 500,000–1 million documents depending on sub-collection.

- Document Level Averages
- Average precision is calculated at specific document cutoffs (e.g., after 5, 10, 20, or 100 relevant documents), not standard recall levels. Average R-precision (precision when R relevant documents have been found) might also be reported.
- Average Precision Histogram
- A graph displays a single precision measure for each topic, generally comparing R-precision between a specific retrieval system and the

system-wide average for each topic.

- Four main evaluation measures are used at the TREC conferences: summary table statistics, recall-precision averages, document level averages, and average precision histograms.
- Summary Table Statistics
- These tables show statistics such as the number of topics, documents retrieved, relevant documents retrieved, and the total number of relevant documents that could be retrieved for all topics.
- Recall-Precision Averages
- This involves tables or graphs of average precision (across all topics) at 11 standard recall levels, using interpolation when actual recall levels do not match standard levels. Non-interpolated average precision may also be included.
- **Purpose of TREC**

- Provide a **benchmark environment** for IR experiments.
- Allow researchers to evaluate and compare different techniques (e.g., automatic thesauri, term weighting, NLP, relevance feedback, pattern matching).
- Encourage reproducible, large-scale IR research.

• The CACM Collection contains all 3,204 articles from Communications of the ACM from 1958 to 1979, serving as a major resource for computer science literature. • Content and Structure

- The collection includes both article text and structured subfields ("concepts"), such as author names, date information, word stems from titles/abstracts, hierarchical categories, direct references, bibliographic coupling, and co-citation counts.

• Word stems are produced by stemming terms from each article’s title and abstract. • Categories come from the Computing Reviews scheme, with most articles having fewer than five categories.

- Direct references are pairs of articles where one cites another.
- Coupling and co-citation data detail connections via shared references or citations, supporting advanced retrieval algorithm testing.

- **Test Information Requests**
- A set of 52 test queries (example: articles discussing IBM's TSS OS) supports information retrieval experiments.
- Each query provides Boolean formulations and a list of relevant articles, typically around 15 per query, making the test set specific and resulting in low average precision/recall statistics.

Keyboard based Querying

- Single Word Queries
- Context Queries
- Boolean Queries
- Natural Language

Keyword-based Querying

- A query is how a **user expresses their information need**, typically using a combination of words.
- This query is then used **to search through a document collection** to find documents that include the same words.
- Word queries are **straightforward and efficient**, as they align with **natural language** and quickly help rank relevant documents.
- **Single word, multiple word and pattern based queries** are classiffied as **Basic Queries**



























































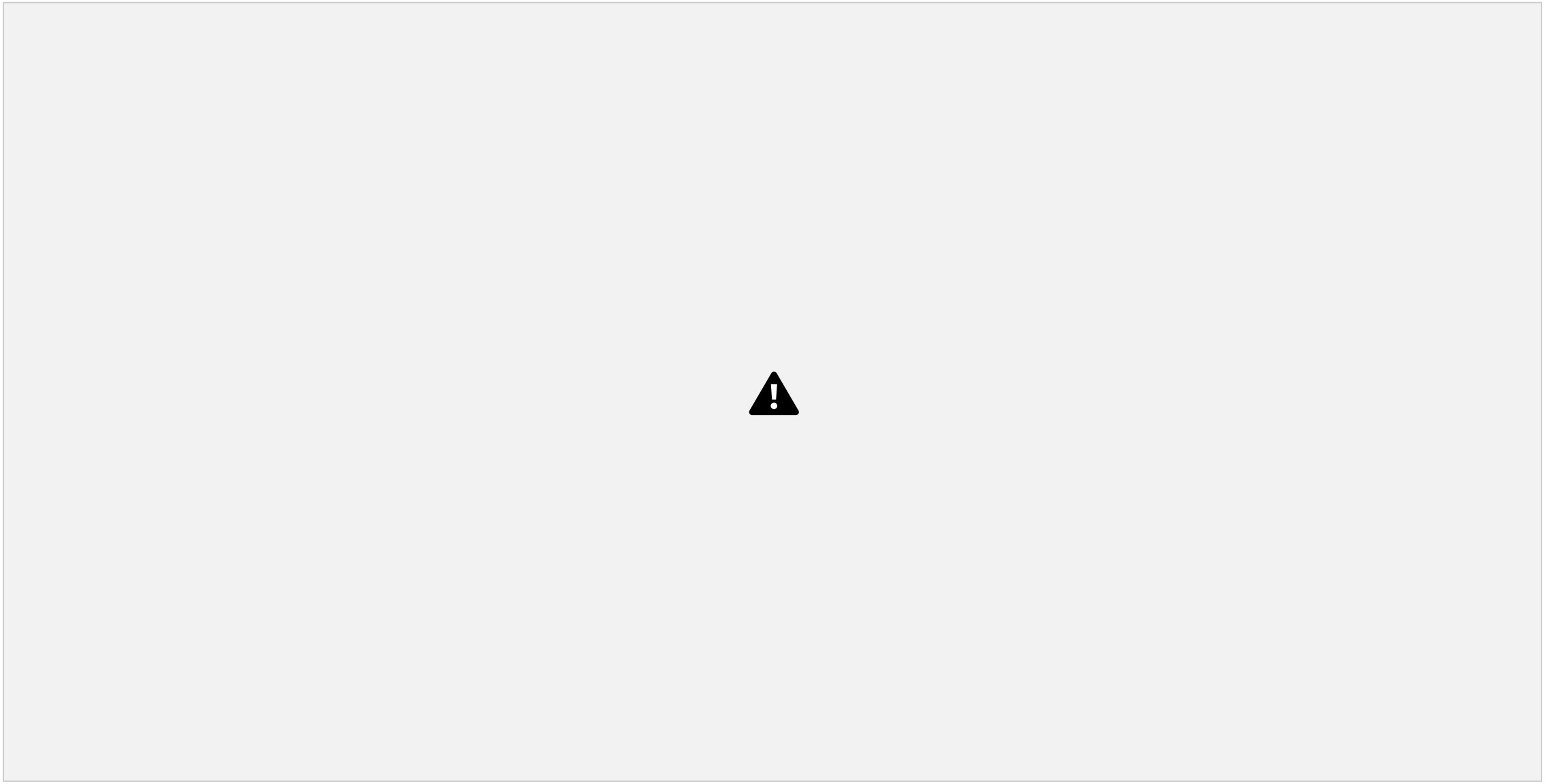












Query protocols







